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Positioning and Connection of Devices in the Vertical Form Fill Seal Machine

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1. Component Location

This document explains the physical location of the main components of the Vertical Form Fill Seal Machine.

The goal is to show how the key elements are distributed and positioned within the system, making it easier to understand for assembly and operation.

Throughout this document, several screenshots of the machine's 3D model will be included, highlighting the important points.

The Vertical Form Fill Seal Machine is equipped with the components listed in the table below:

Units	Device	Device Description	Note
1	Servo Cross Sealer Lift Axis	SINAMICS S200 PN	Servo used to control the vertical movement of the cross-sealing unit
1	Servo Filling Axis	SINAMICS S200 PN	Servo used to feed material into the machine
1	Servo Foil Feed Axis	SINAMICS S200 PN	Servo used to push the film
1	Unwinder motor		Motor used to unwind the film based on the tension control
1	Cross sealer jaw	Air cylinder	Cylinder used to seal and cut products in the horizontal direction. It is installed on the Cross Sealer Lift Axis
1	Vertical sealer jaw	Air cylinder	Cylinder used to seal the film and shape it into a tubular form.
1	Analog sensor for tension control	Generic analog sensor	Sensor used to measure and acquire the tension level of the foil
1	Digital color code sensor	Generic digital sensor	Digital sensor for detecting color marks

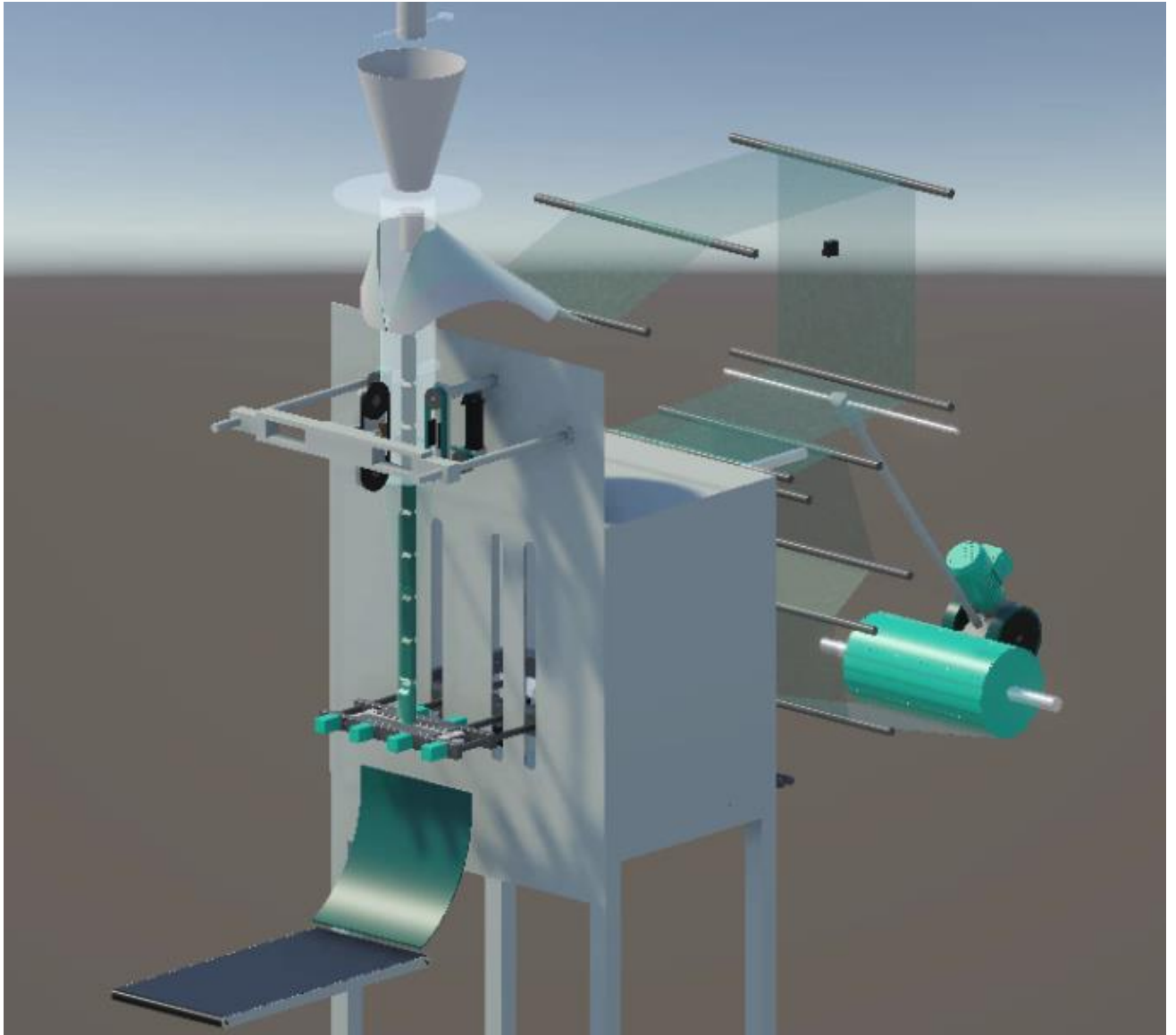


Figure 1-1: Vertical Form Fill Seal Machine

The positioning of the listed components is as detailed below:

Servo Cross Sealer Lift Axis → Horizontal servo that moves the sealing jaws of the machine up and down. It uses a belt system to create this vertical movement, helping the machine seal the film at the right time during the packaging process.

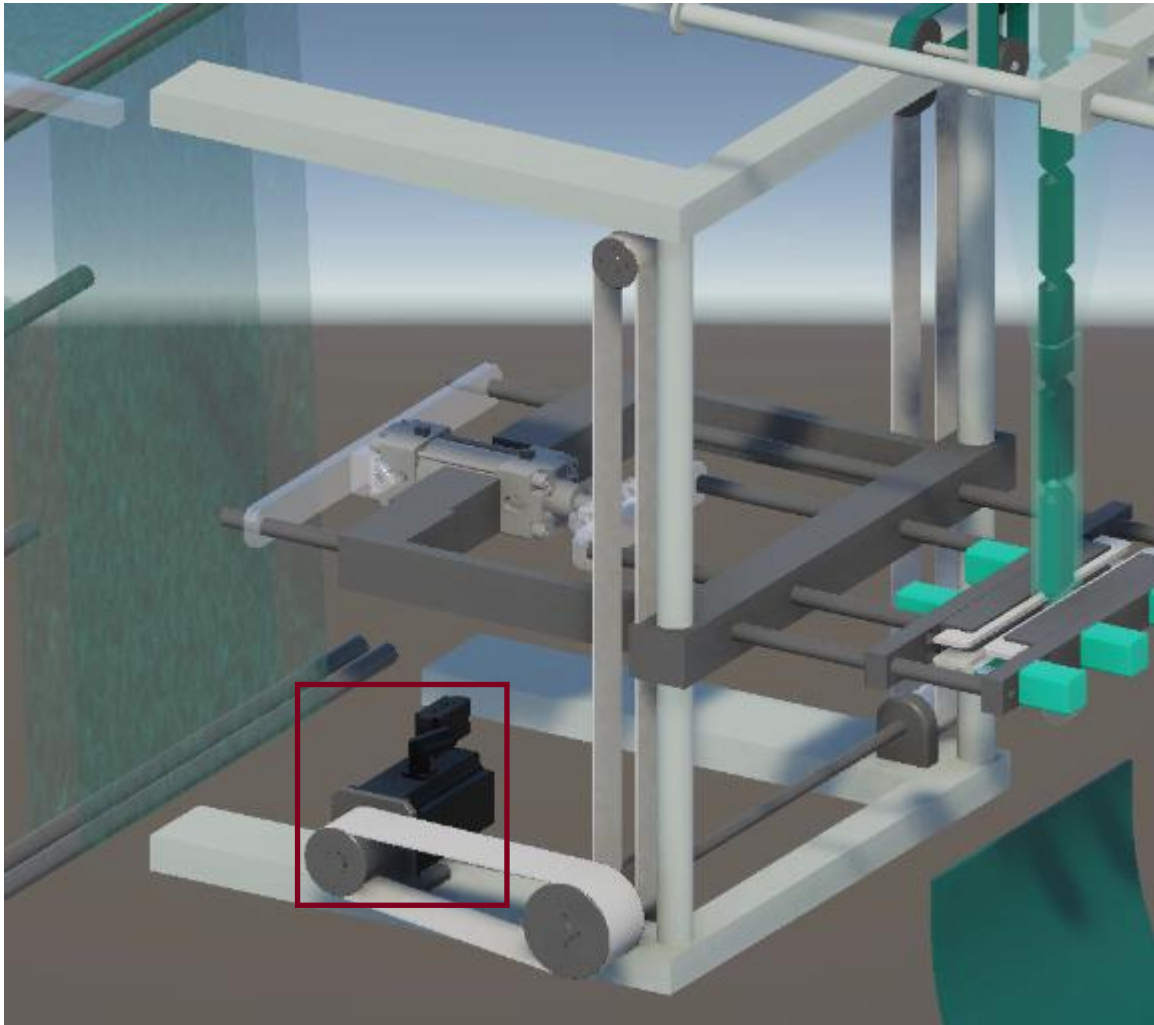


Figure 1-2: Cross Sealer Unit

Servo Filling Axis → The Servo Filling Axis is a servo located over the feeding cone. It is responsible for feeding the product that will be packed during each cycle.

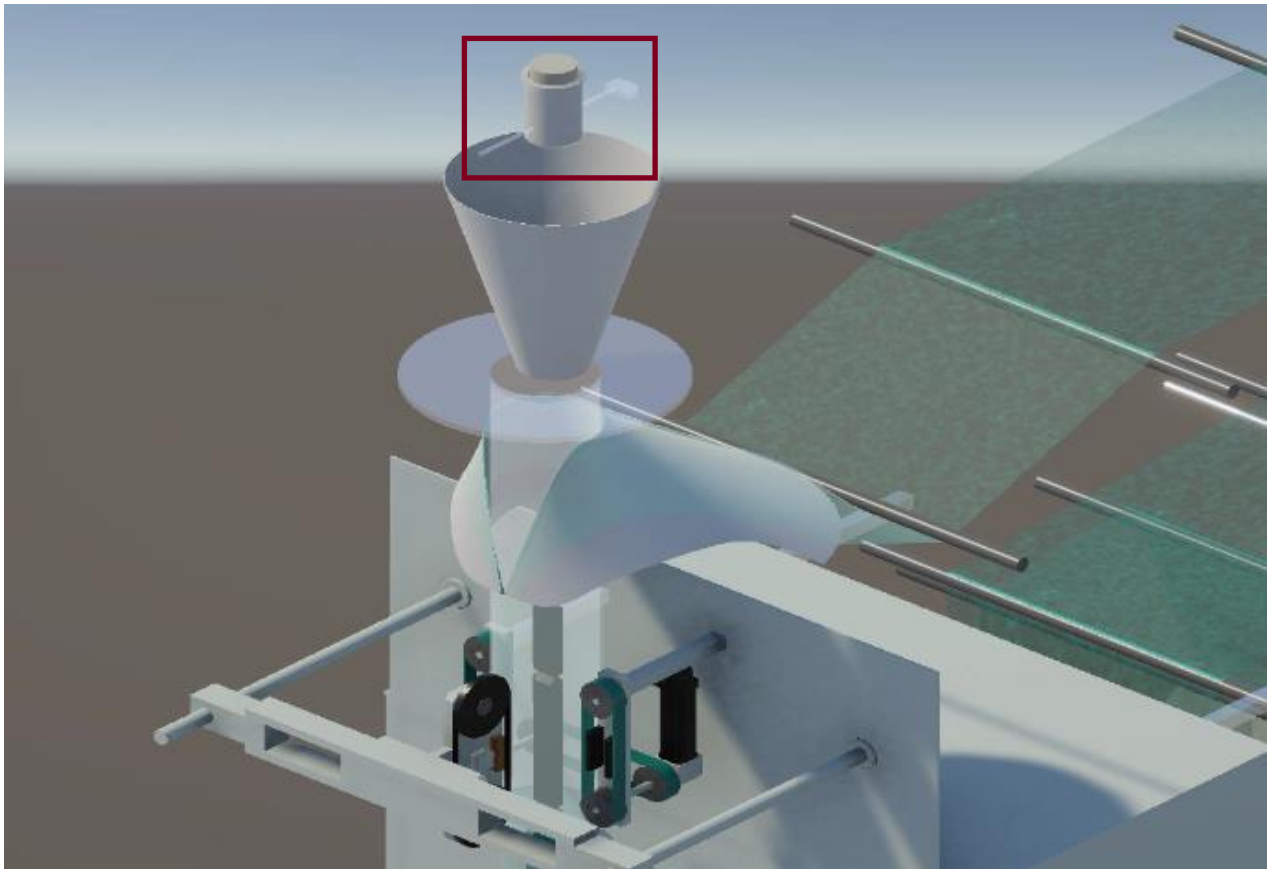


Figure 1-3: Filling System

Servo Foil Feed Axis → Servo positioned vertically below the feeding area, used to push the film forward through belts connected to the servo.

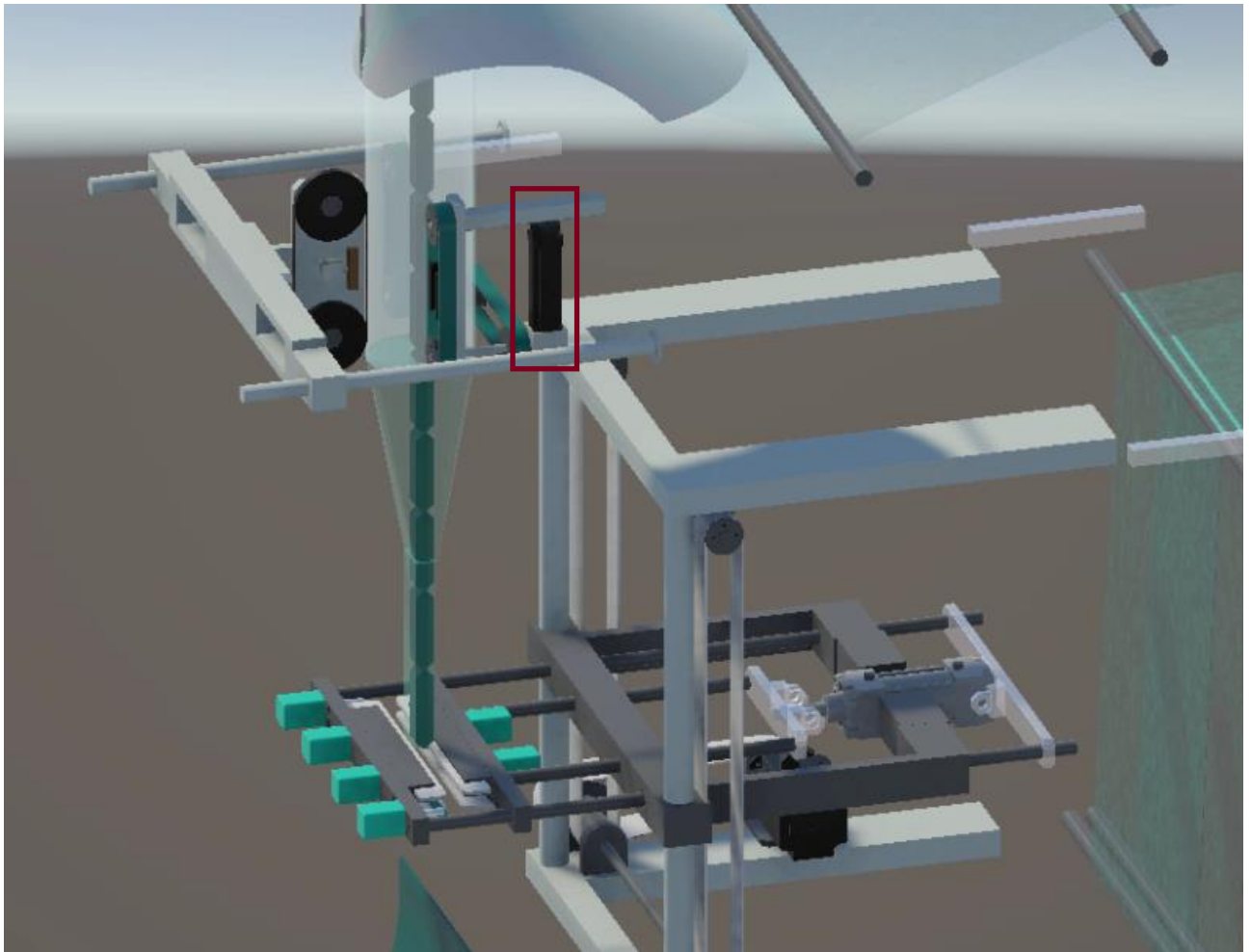


Figure 1-4: Foil Feed Axis

Unwinder motor → Motor positioned at the back of the machine, in direct contact with the film, responsible for rolling and unrolling the film based on tension control.

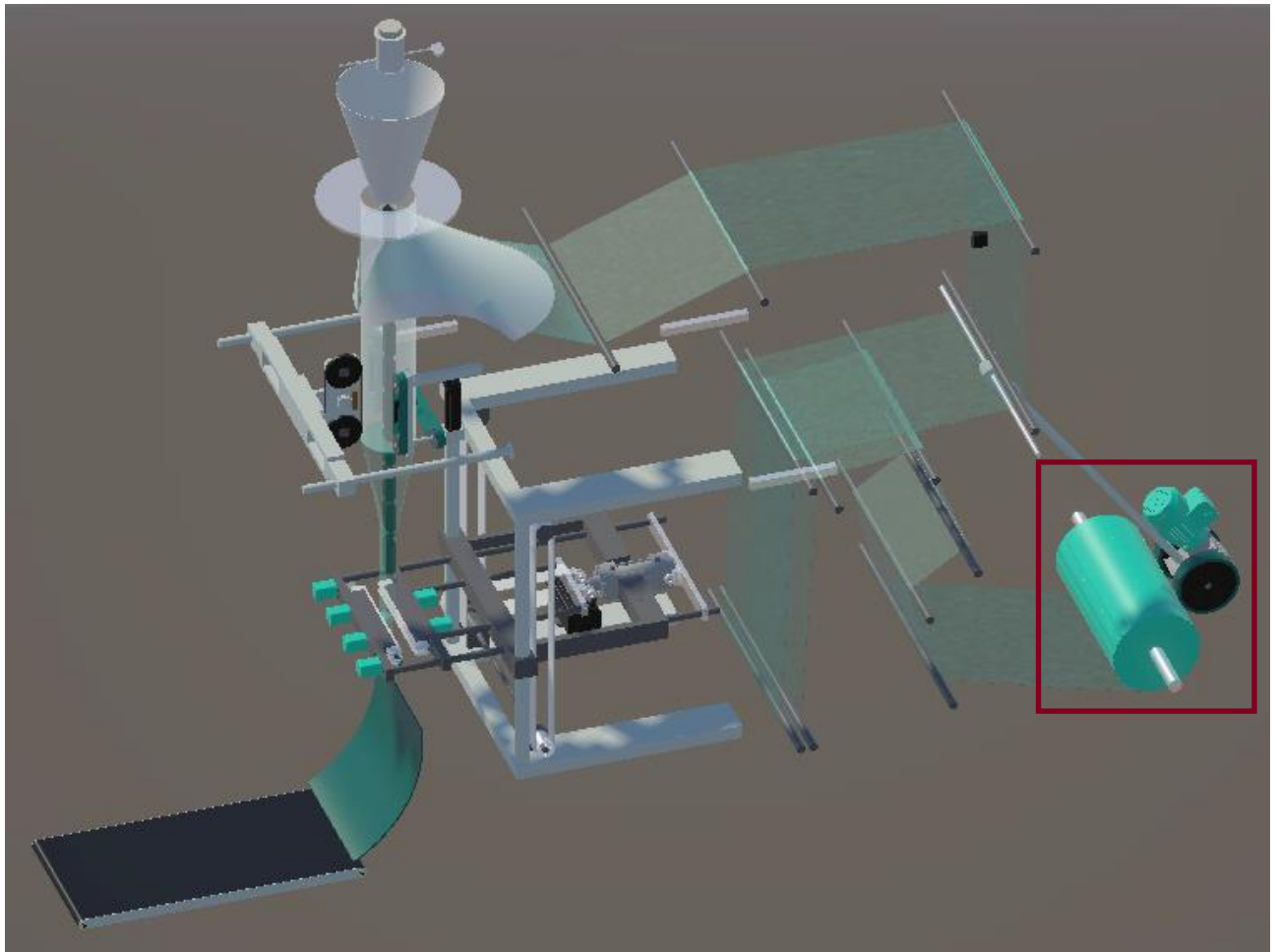


Figure 1-5: Unwinding mechanism

Analog sensor for tension control → Tension sensor mounted above the film, responsible for continuously measuring the film tension and transmitting data to the tension control system to maintain stable film transport.

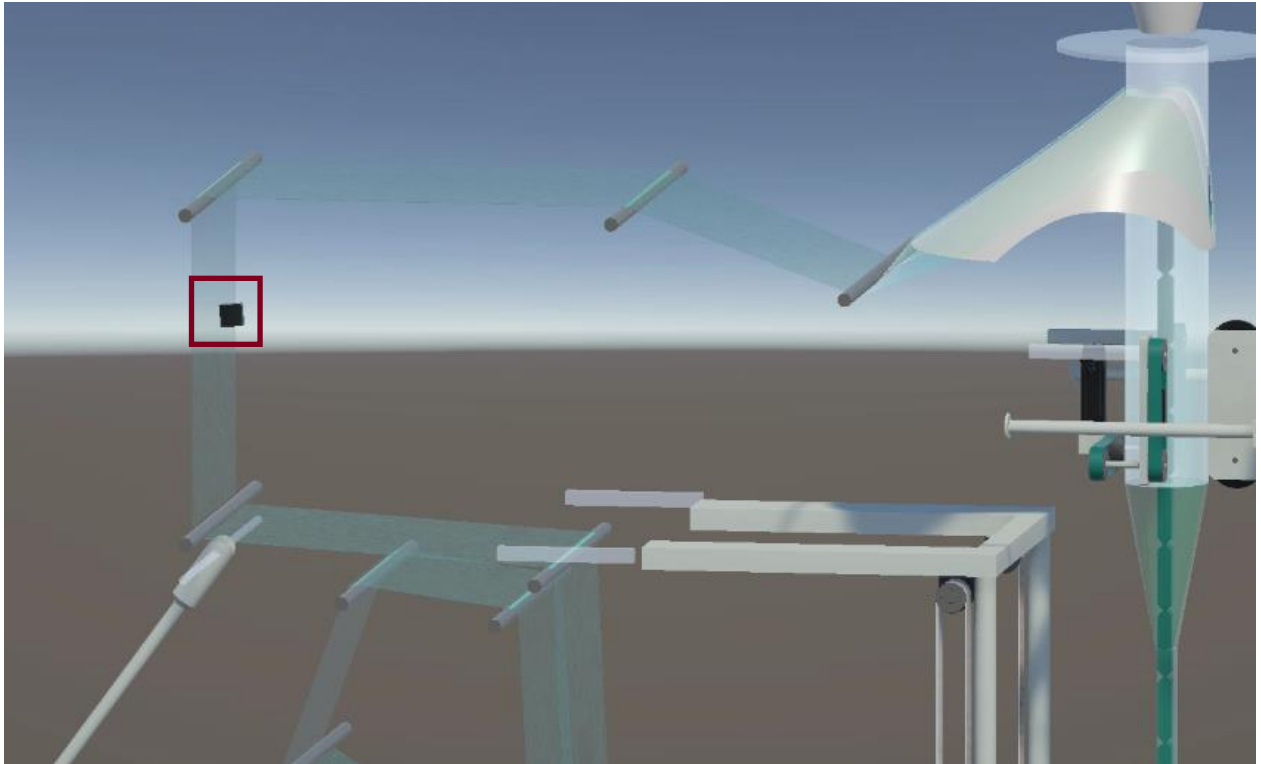


Figure 1-6: Basic Tension Control

Cross sealer jaw → Air cylinder mounted on the Cross Sealer Lift Axis, which enables their vertical movement. The Cross sealer jaw is responsible for executing the horizontal sealing motion required to close and cut the packaging film during each cycle.

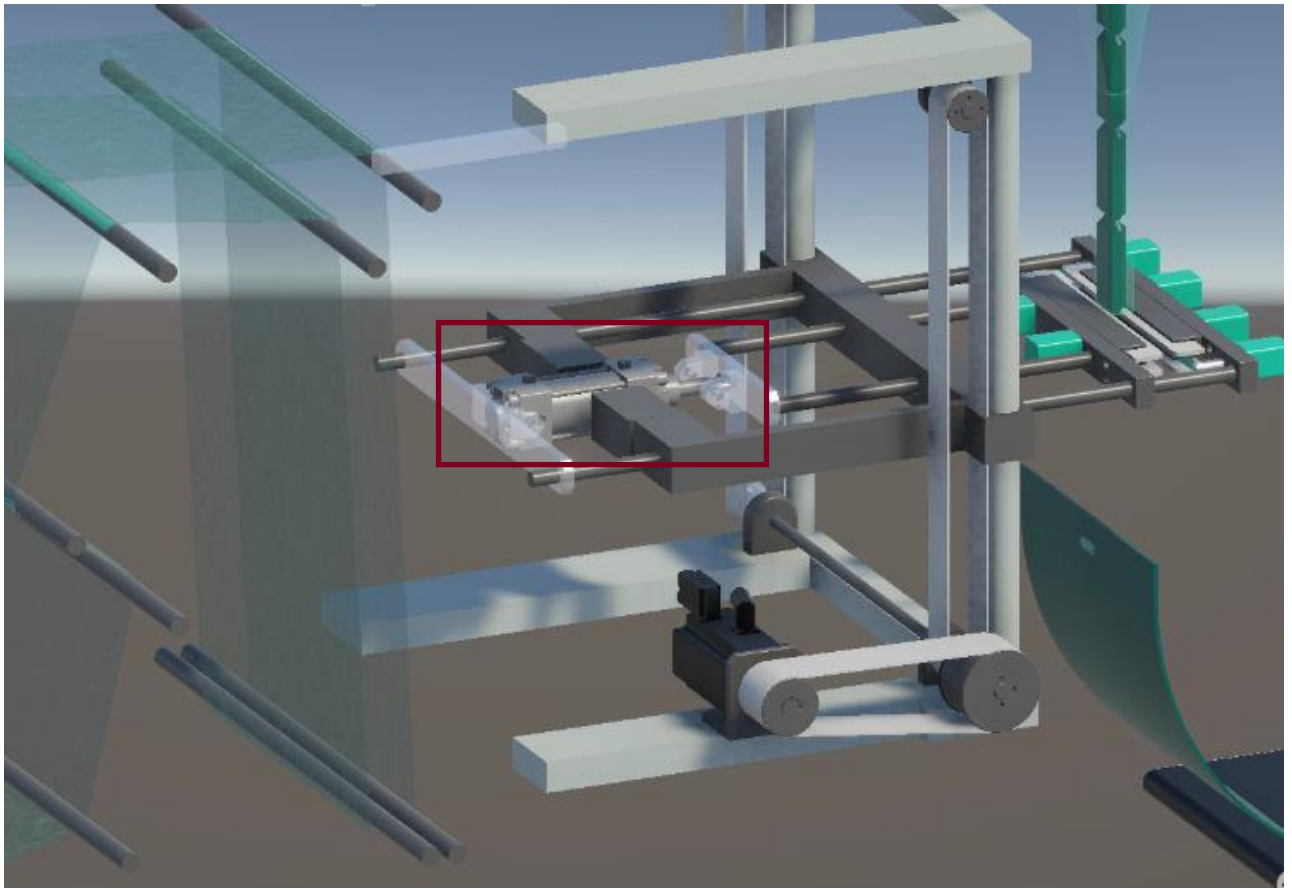


Figure 1-7: Cross Sealer Jaw

Vertical sealer jaw → Vertical sealer jaw positioned vertically below the feeding area, similar to the Servo Foil Feed Axis. It is used to seal the film and form it into a tubular shape. The jaw extends to the working position only while the film is moving, ensuring synchronized and precise sealing during continuous operation.

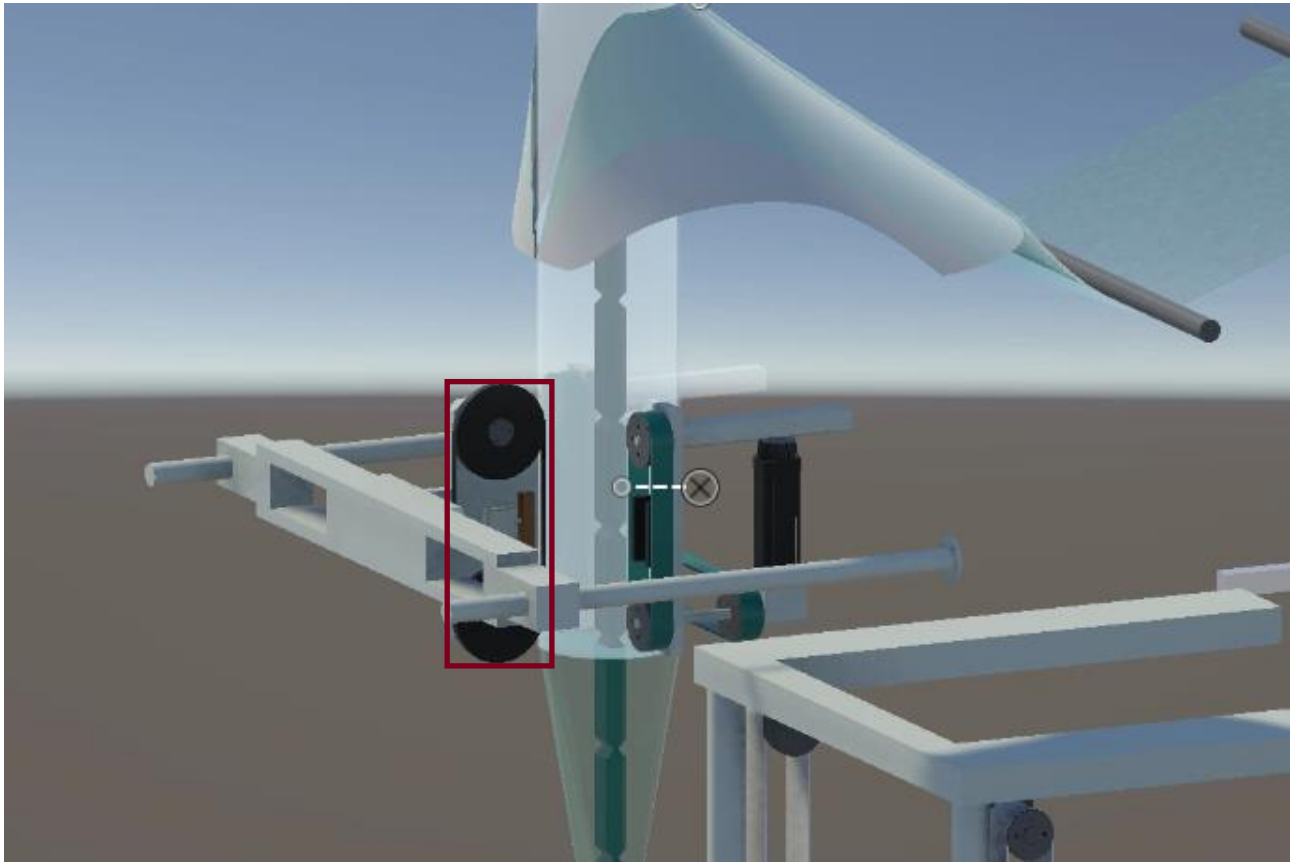


Figure 1-8: Vertical Sealer Jaw

2. Electrical connections

This section of the document includes basic electrical drawings for setting up the Vertical Form Fill Seal Machine:

- Drive Connection:

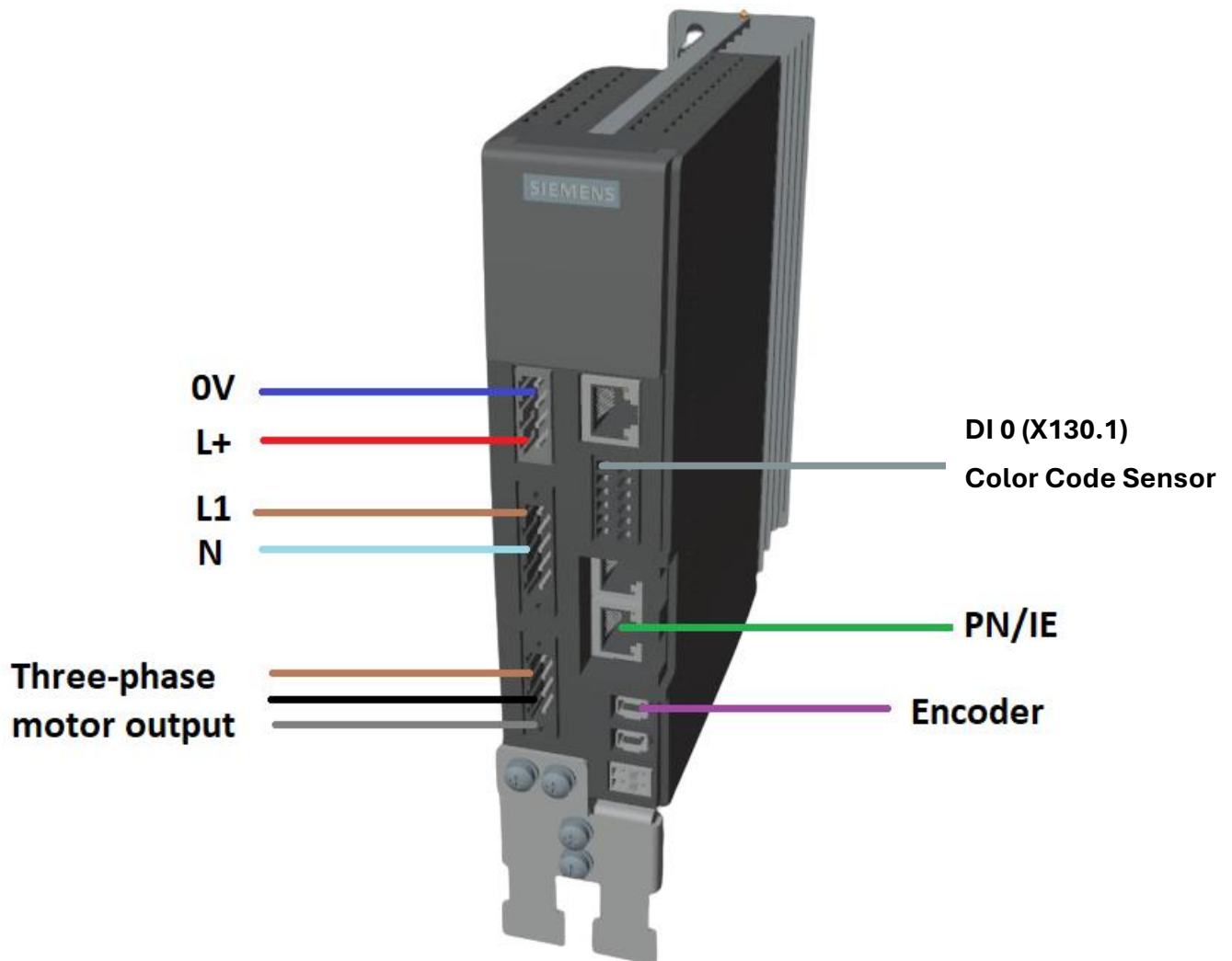
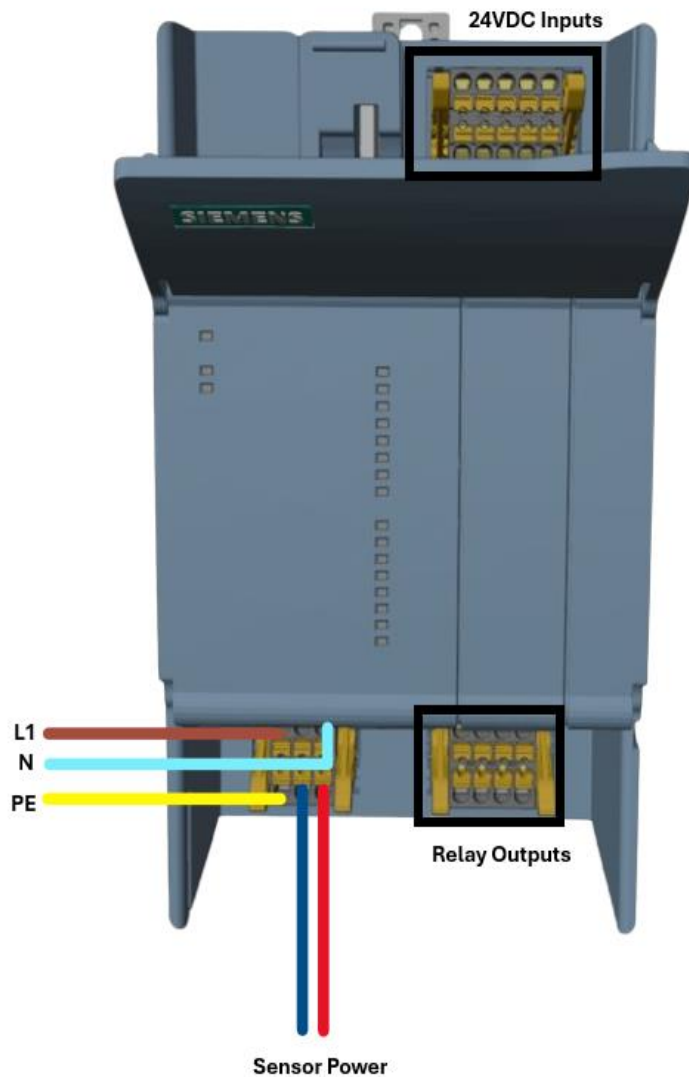


Figure 2-1: Drive Connection

It is important to note that in this machine application, the "Color Code Sensor" is connected directly to the "Foil Feed Axis" drive rather than to the PLC. This setup offers several advantages: it eliminates the need for a technology module to perform the control, reduces the PLC's processing load and provides more precise and faster synchronization with the motor.

- PLC SIMATIC S7-1200 G2 Connection:



%I0.0 → Vertical Sealer Jaw In Home Position
 %I0.1 → Vertical Sealer Jaw In Work Position
 %I0.2 → Cross Sealer Jaw In Home Position
 %I0.3 → Cross Sealer Jaw In Work Position

%Q0.6 → Vertical Sealer Jaw to Home Position
 %I0.1 → Vertical Sealer Jaw to Work Position Sensor
 %I0.2 → Cross Sealer Jaw to Home Position Sensor
 %I0.3 → Cross Sealer Jaw to Work Position Sensor

Figure 2-2: PLC SIMATIC S7-1200 G2 Connection

- Line Connections:

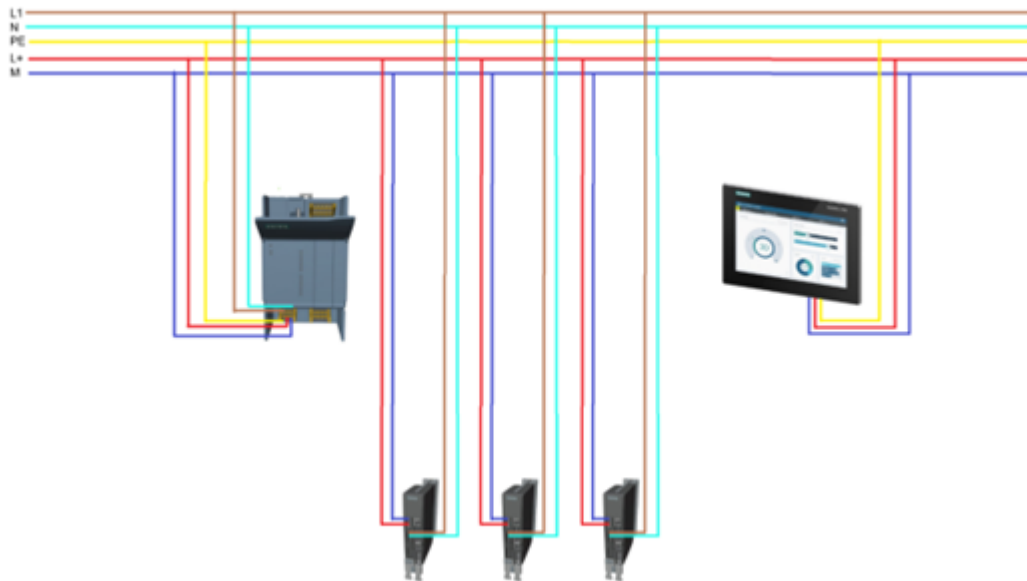


Figure 2-3: Line Connection

- Network Connection:

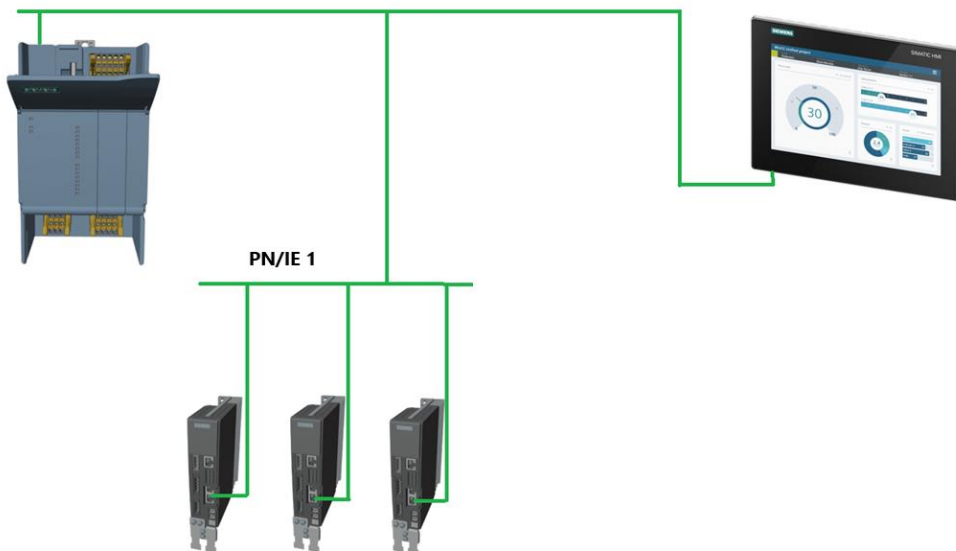


Figure 2-4: Network Connection